

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1315

South Colton 138 kV
February 21, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1315. The Interconnection Customer proposes a 105.8 MW battery energy storage facility interconnecting at the South Colton 138 kV substation in the Virginia area. The request was received on October 22, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 443 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$25,889,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Grassmere to Red Hill 138 KV L	Clover to Bear Substation 345 KV	120.1	295	5.6
Boonsboro to Reusens 138 KV	Clover Substation to Bear 345 KV	118.6	388	5.2
Coal Mtn to Grassmere 138 KV	Bear to Clover 345 KV Line	117.4	329	5.6
Coal Mtn to Central Austin 138 KV line	Bear to Clover 345 KV Line	116.7	338	5.6
Central Austin to Waynesboro 138 KV Line	Bear to Clover 345 KV	116.1	348	5.6
Waynesboro to Alderwood 138 KV	Bear to Clover 345 KV	114.4	377	5.6
Walworth to Shenandoah 138 KV	Bear to Clover 345 KV	111.5	345	5.5
Shenandoah to Aibonito Pds 138 KV line	Bear to Clover 345 KV	110.5	367	5.5
Monel to Candelaria 138 KV	Bear to Clover Substation 345 KV	107.7	443	8.2
Bear to Fluvanna P.S 345 KV L	Bear to Leavitt 345 KV #1	106.5	1,358	13.3
Hinton Ap to Oxnard 138 KV	Coal to McClain Energy Facility 138 KV Line	104.0	229	5.2
De Kalb to Oxnard 138 KV	Coal to McClain Energy Facility 138 KV	103.8	223	5.2
McClain Energy Facility to De Kalb 138 KV	McClain Energy Facility to Coal 138 KV	102.3	220	5.2
Bear to Mecklenburg Power Station 345 KV L	Bear to Clover Substation 345 KV	101.2	2,827	19.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Grassmere to Red Hill 138 KV L	Rebuild line with higher-capacity conductor	\$2,110,000	27% (\$561,000)
Reusens to Boonsboro 138 KV	Rebuild line with higher-capacity conductor	\$1,814,000	7% (\$123,000)
Coal Mtn to Grassmere 138 KV	Rebuild line with higher-capacity conductor	\$5,519,000	27% (\$1,467,000)
Coal Mtn to Central Austin 138 KV line	Rebuild line with higher-capacity conductor	\$5,454,000	27% (\$1,450,000)
Waynesboro to Central Austin 138 KV	Rebuild line with higher-capacity conductor	\$3,099,000	27% (\$824,000)
Waynesboro to Alderwood 138 KV	Rebuild line with higher-capacity conductor	\$3,948,000	27% (\$1,050,000)
Shenandoah to Walworth Substation 138 KV	Rebuild line with higher-capacity conductor	\$23,718,000	28% (\$6,601,000)
Aibonito Pds to Shenandoah 138 KV	Rebuild line with higher-capacity conductor	\$35,366,000	28% (\$9,843,000)
Monel to Candelria 138 KV	Reconductor line	\$12,066,000	17% (\$2,038,000)
Bear Substation to Fluvanna P.S 345 KV line	Reconductor line	\$48,548,000	1% (\$573,000)
Oxnard to Hinton Ap 138 KV Line	Reconductor line	\$1,654,000	6% (\$101,000)
De Kalb to Oxnard 138 KV	Reconductor line	\$3,763,000	6% (\$229,000)
Mcclain Energy Facility to De Kalb 138 KV	Reconductor line	\$1,420,000	6% (\$87,000)
Bear to Mecklenburg Power Station 345 KV L	Reconductor line	\$63,589,000	2% (\$942,000)
	Total allocated to project		\$25,889,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.